Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	ERROR 303
Project / Solution Name	StockWise
Name (Leader)	Timothy Lai Yong Fong
Email & Phone (Leader)	timothyfy851@gmail.com / +60 111325 8211
University Affiliation(s)	Taylor's University
Live Prototype URL	https://stock-wise.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/timlai0402/ERROR-303.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Component	Proposed Technology	Purpose
Frontend	HTML	Provides an interactive dashboard for managing inventory, orders, suppliers, and restocking recommendations.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
HTML5	Structure of the web application	NA

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
NA	NA	NA

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Cloud IDE, workspace development, prototype layout construction, and live staging	Starter Tier
GitHub Pages	Production static web application edge hosting and continuous integration	Free Tier

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
NA	NA	NA	NA

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic customer orders and bills of materials	Generated by the team; no external URL	Sysnthetic	NA	N
synthetic inventory data	Generated by the team; no external URL	Synthetic	NA	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

- ☑ Yes
- ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit AI Assistant	Real-time code completions, layout debugging, and syntax checks	Major
Gemini Argon 4	Provide recommendation to elevate our project idea	Moderate
Codex	Refine the code	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

All AI-suggested code, styling components, and copy underwent manual peer review and validation. Generated code was refactored to align with React 19 standards and security best practices, eliminating unnecessary dependencies or exposed parameters. No confidential project keys or real business data were exposed to external prompts. Team ERROR 303 assumes complete responsibility for the architecture, software stability, and accuracy of all code in the repository.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

- ☑ Yes
- ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Cloud IDE platform and application hosting environment	Scaffolded the first-edition UI layout and developed component logic

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was used as an agile workspace to build, iterate, and preview the web interface. While it expedited development and testing, it is not strictly essential to the core functionality of StockWise AI. The solution can be hosted on standard deployment platforms such as Github Pages or Vercel.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

StockWise AI collects and stores zero personal, customer, or real company data in its current prototype. The application runs entirely on team-generated synthetic data modeling stock quantities, unit costs, supplier names, and product bills of materials. In the proposed production Phase 2 deployment, data stored in Supabase PostgreSQL will be encrypted in transit via TLS 1.3 and at rest via AES-256. In accordance with Malaysia's Personal Data Protection Act (PDPA 2010), role-based access control (RBAC) ensures operators only access role-relevant data. External AI API calls will receive strictly anonymized material parameters, ensuring no identifiable proprietary or personal records are transmitted or stored externally

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The primary technical risks include:
(1) GenAI hallucination or non-deterministic outputs when assessing procurement risk;
(2) third-party API downtime or rate-limiting; and
(3) stale external market data.
To mitigate hallucination risks, StockWise AI restricts GenAI exclusively to qualitative summaries and contextual justifications. All quantitative calculations (inventory balances, shortage projections, bill-of-materials deductions, and minimum reorder quantities) are performed by deterministic, rule-based algorithms. Crucially, to prevent unvetted automated spending, the system enforces a strict human-in-the-loop requirement: no purchase order can be dispatched without explicit manual approval from an authorized manager.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



TIMOTHY LAI YONG FONG
24/10/2026